INCH-POUND

MIL-DTL-24558/30B
3 October 2011
SUPERSEDING
MIL-DTL-24558/30A
w/AMENDMENT 1
15 July 2005

DETAIL SPECIFICATION SHEET

TERMINAL BOXES, CONNECTION, SUBMERSIBLE (15-FOOT), 500 VOLTS, THREE-PHASE, SYMBOL NO. 576.1 - TWO CIRCUITS FOR 150 MCM CABLE, ONE CIRCUIT FOR 200 MCM CABLE AND ONE CIRCUIT FOR 400 MCM CABLE, SYMBOL NO. 577.1 - ONE CIRCUIT FOR 150 MCM CABLE, ONE CIRCUIT FOR 200 MCM CABLE AND TWO CIRCUITS FOR 250 MCM CABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification sheet and MIL-DTL-24558, "Terminal Boxes, Connection, for Electrical and Electronic Systems, General Specification for".

REQUIREMENTS: The terminal boxes shall be as specified herein and in accordance with table I and figures 1 through 8.

1. Electrical rating. The terminal boxes shall have an electrical rating of 500 volts root mean square, and 500 volts direct current. Terminal lugs are designed for 150 thousand circular mils (MCM), 200 MCM, 250 MCM, and 400 MCM cables.

2. Dimensions and configuration. For dimensions and configuration see figures 1 through 8. All dimensions in figures 1 through 8 are expressed in decimal inches. Overall terminal box dimensions are approximate. Exact dimensions are shown in MIL-E-24142/7, "Enclosures, Submersible (15-Foot), Size 20 by 24".

3. Tolerance. All tolerances not specified herein shall be in accordance with MIL-DTL-24558.

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4. Standard parts. The standard parts shown in table I shall be used in the construction of the connection boxes specified herein. The numbers in circles shown on figures 1, 2, 4, 5, 6, 7, and 8 are the same as those shown in the "Item" column of table I.

5. Enclosure. The enclosures shall be a 15-foot submersible type.

6. Finish. The interior of the enclosures may be painted or left unpainted at the contractor's option.

7. Weight. The weight of the connection box shall not exceed 81.4 pounds.

8. Lugs. The contact surface of the terminal lugs, items 6, 7, 8, and 21 as shown in table I, and figure 5 shall be silver plated. The plating thickness shall be not less than 0.0005 inch.

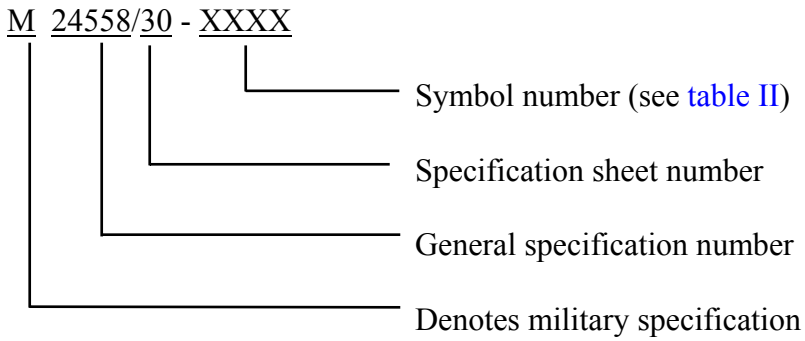
9. Terminal box. Each terminal box shall be furnished complete with terminal lugs installed as shown for symbol no. 576.1. In order to convert 576.1 to 577.1, add six 250 MCM terminal lugs, item 21 (see table I) to be packaged inside the enclosure.

QUALITY ASSURANCE: First article inspection and quality conformance inspection shall be as specified in MIL-DTL-24558.

NOTES:

1. Drawing. The terminal box specified by this specification sheet was formerly covered on NAVSHIPS drawing no. 9000-S6202-74182.

2. Part or identifying number (PIN). The PIN to be used for terminal boxes acquired to this specification is created as follows:



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TABLE I. Parts list.

Item	Description	Qty	Part number or specification	Remarks
1	Enclosure assembly, brass	1	MIL-E-24142/7	M24142/7-001
2	Identification plate	1	MIL-DTL-24558	Figure 6
3	Information plate	1	MIL-DTL-24558	Figure 6
4	Terminal board base	1	Figure 4	MIL-I-24768/1 type GME
5	Mounting bracket	3	MIL-DTL-24558	Figure 2
6	Terminal lug 150 MCM, solderless	6	Figure 5	ASTM B187/ B187M copper
7	Terminal lug 200 MCM, solderless	3	Figure 5	ASTM B187/ B187M copper
8	Terminal lug 400 MCM, solderless	3	Figure 5	ASTM B187/ B187M copper
9	Terminal bus assembly - phase "A"	1	Figure 6	ASTM B187/ B187M copper
10	Terminal bus assembly - phase "B"	1	Figure 7	ASTM B187/ B187M copper
11	Terminal bus assembly - phase "C"	1	Figure 8	ASTM B187/ B187M copper
12	RDH machine screw 6-32 X 0.250	4	Commercial	Brass
13	RDH machine screw 7/16-14 X 1.188	8	Commercial	Steel
14	Washer, 7/16 internal tooth	8	Commercial	Steel
15	RDH machine screw 3/8-16 X 1.188	18	Commercial	Steel
16	Washer, 3/8 flat	18	Commercial	Steel
17	Stud 3/8-16 X 2.250	12	Commercial	Brass
18	Hex head nut 3/8-16	24	Commercial	Brass
19	Washer, 3/8 internal tooth	24	Commercial	Brass
20	Pin 3/32 diameter - length to suit	12	ASTM B124/ B124M	Copper
21	Terminal lug 250 MCM, solderless	6	Figure 5	ASTM B187/ B187M copper

TABLE II. Symbol numbers and general information.

Symbol number	For cable size MCM [1]
576.1	150, 200, 400
577.1	150, 200, 250

[1] MCM is defined as thousand circular mils.

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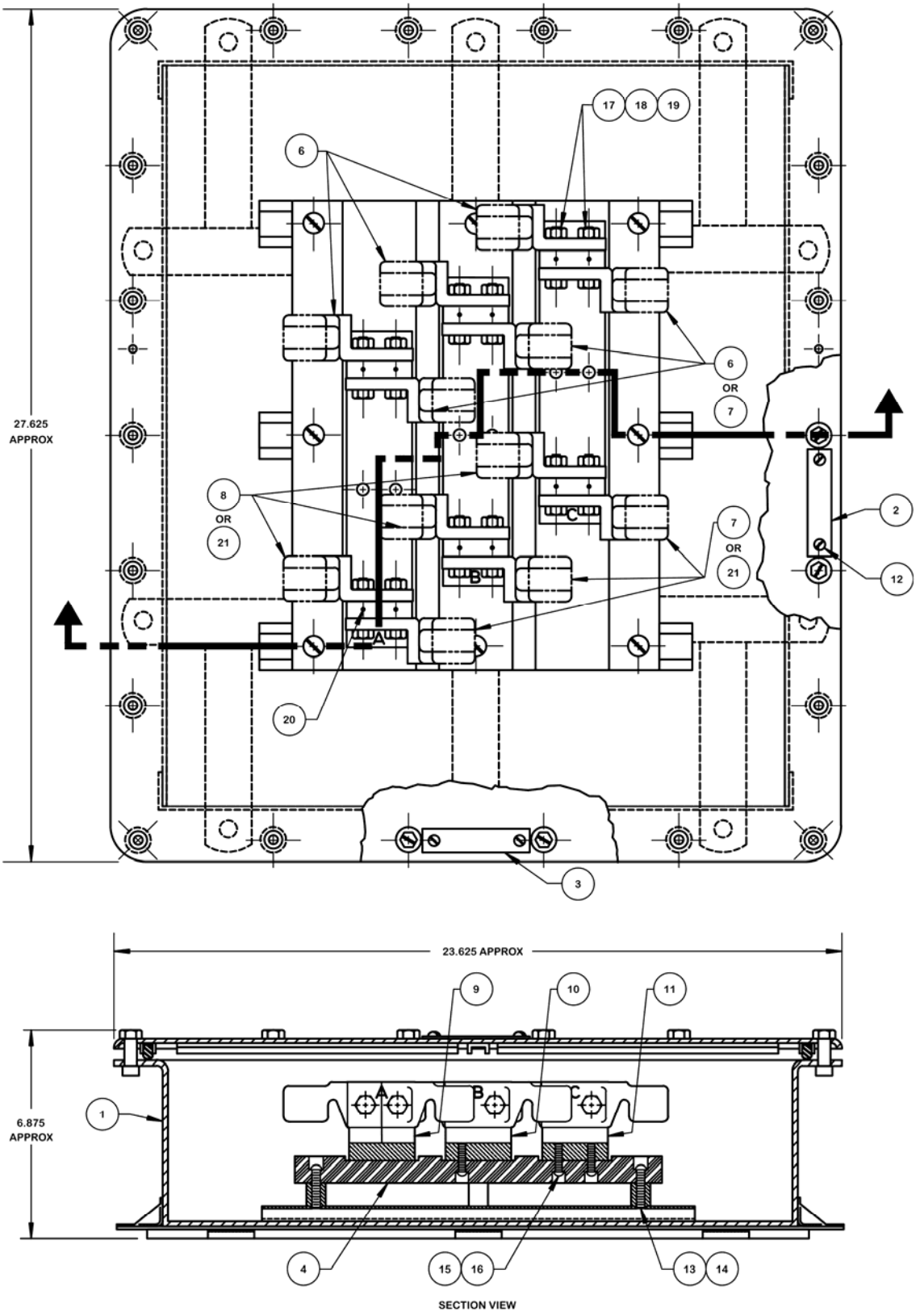


FIGURE 1. Connection box assembly.

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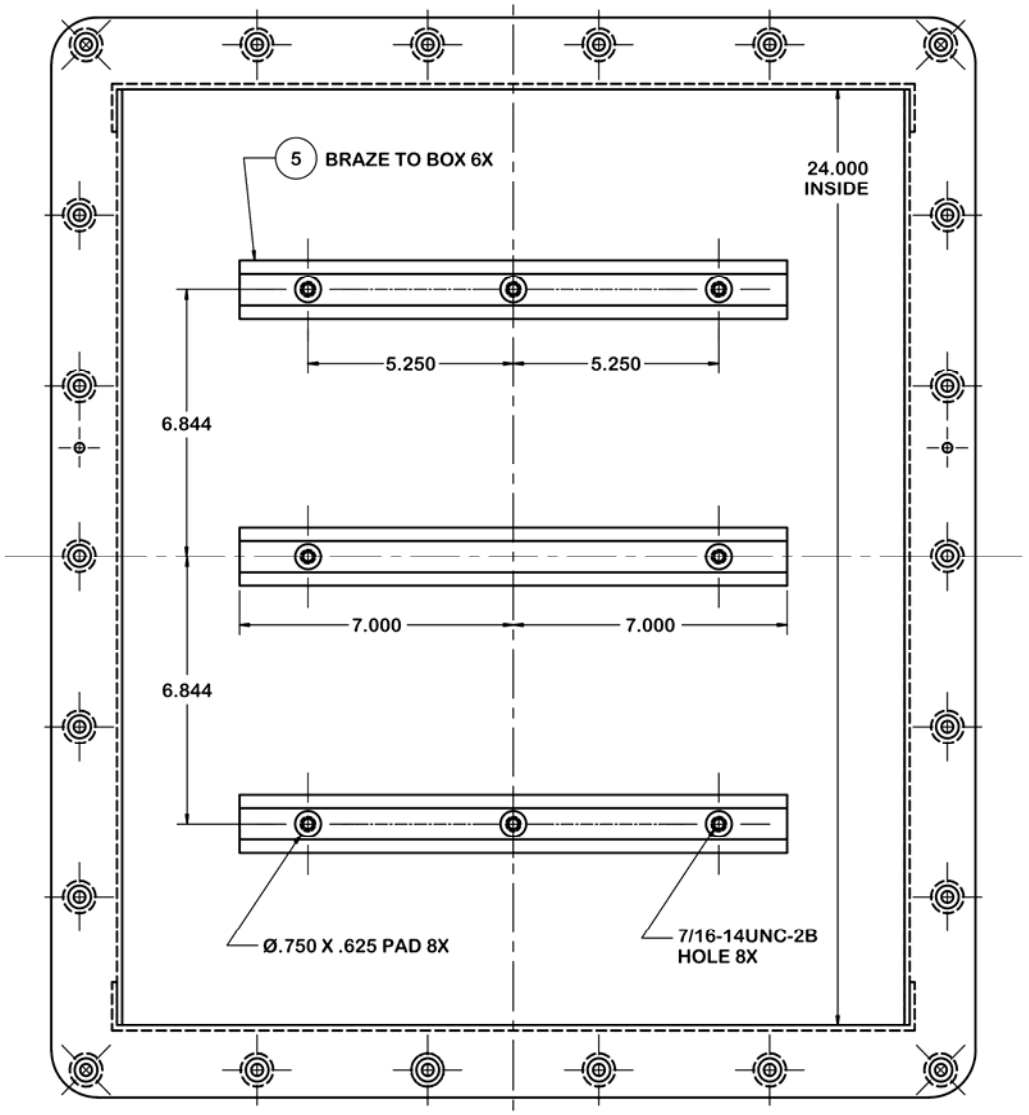


FIGURE 2. Location of terminal board mounting brackets.

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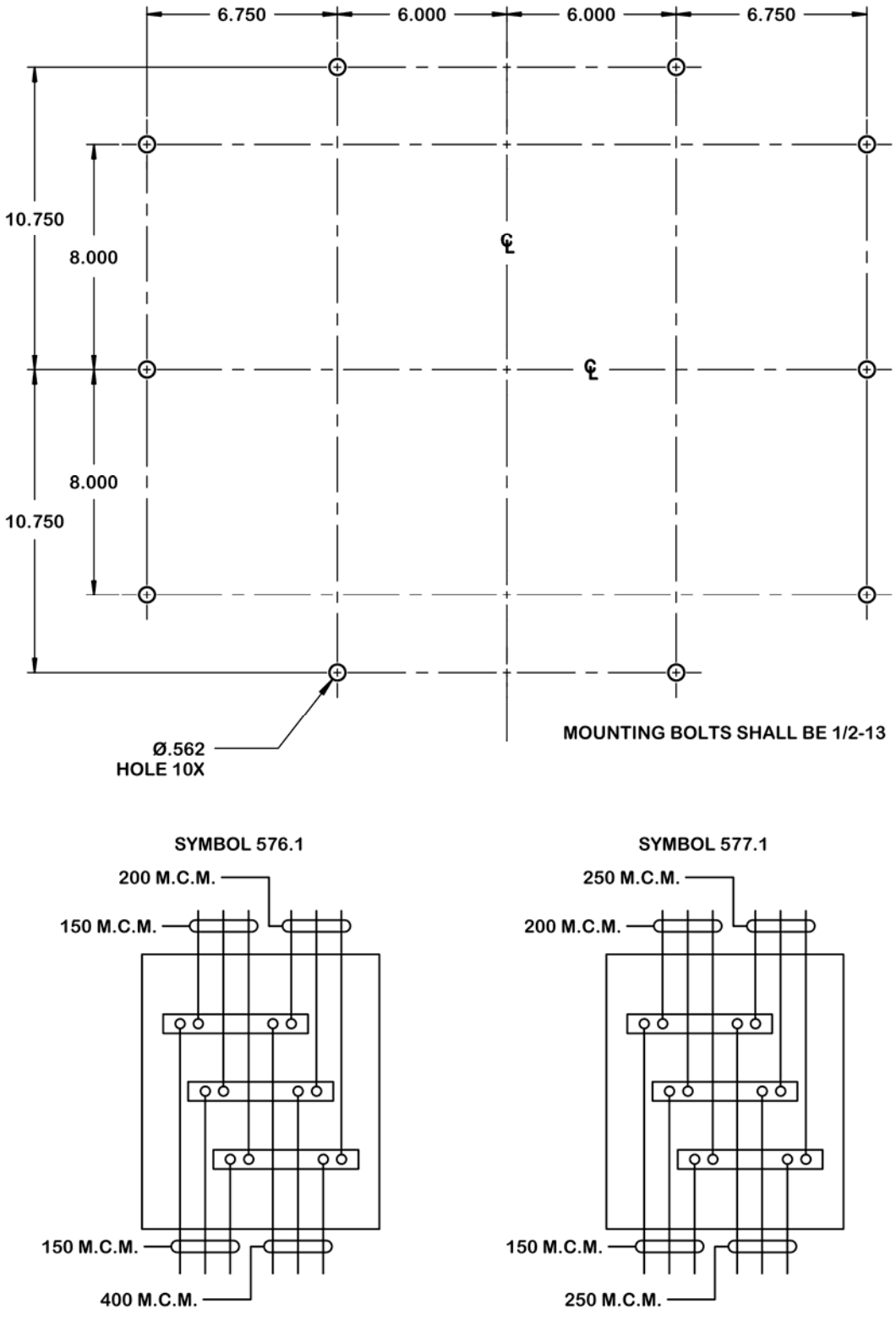


FIGURE 3. Mounting diagram and wiring diagrams.

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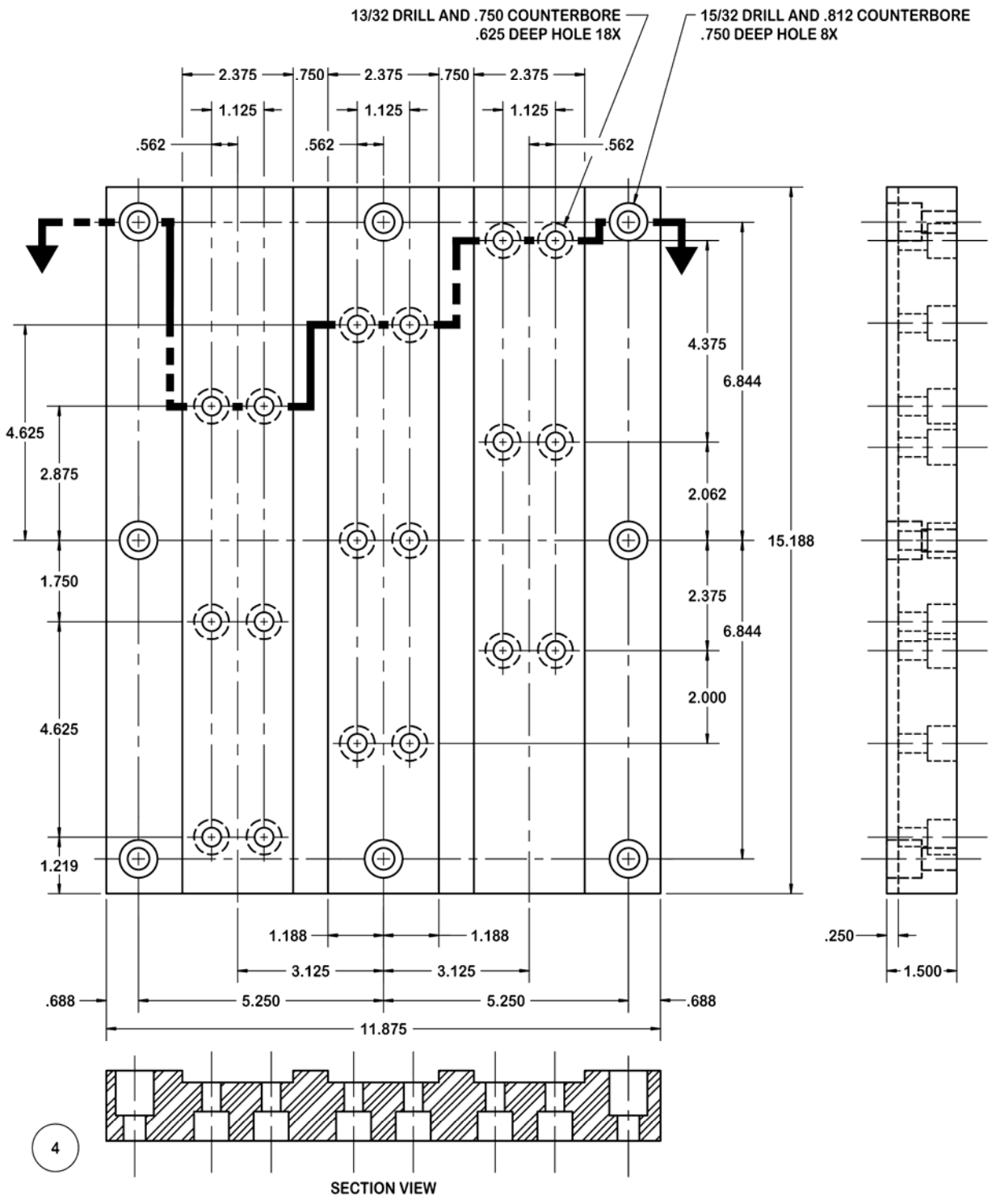


FIGURE 4. Terminal board base.

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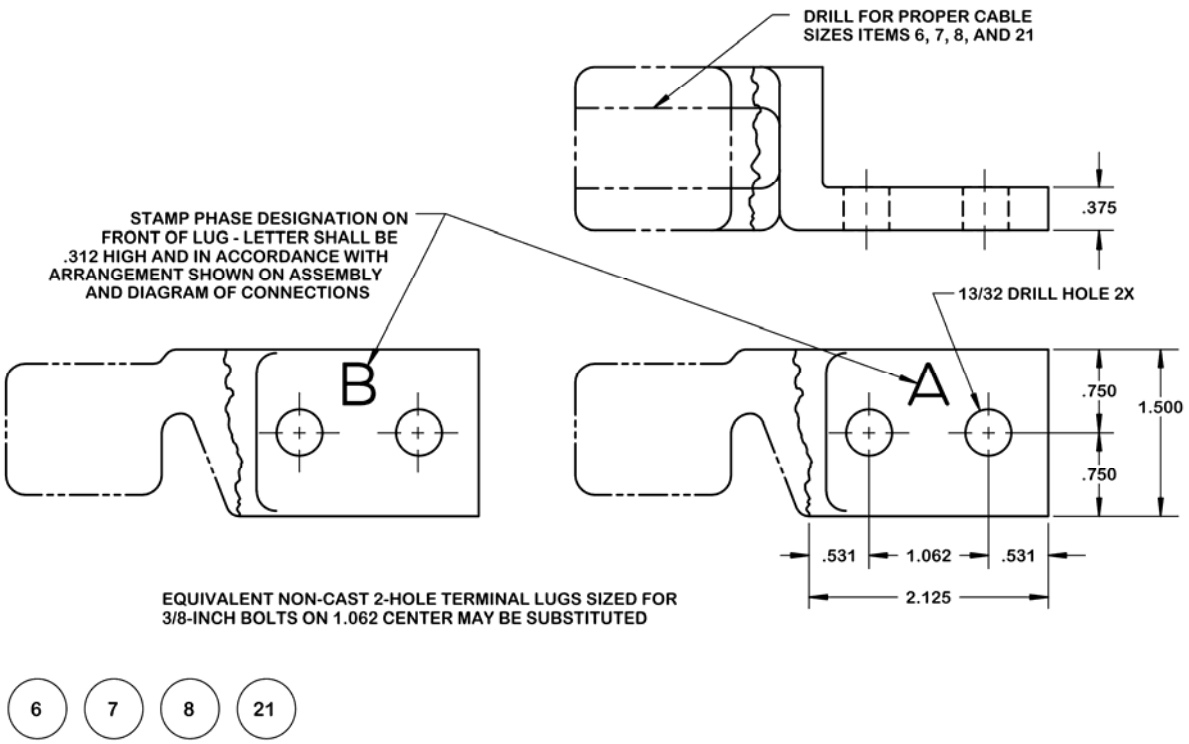


FIGURE 5. Terminal lugs.

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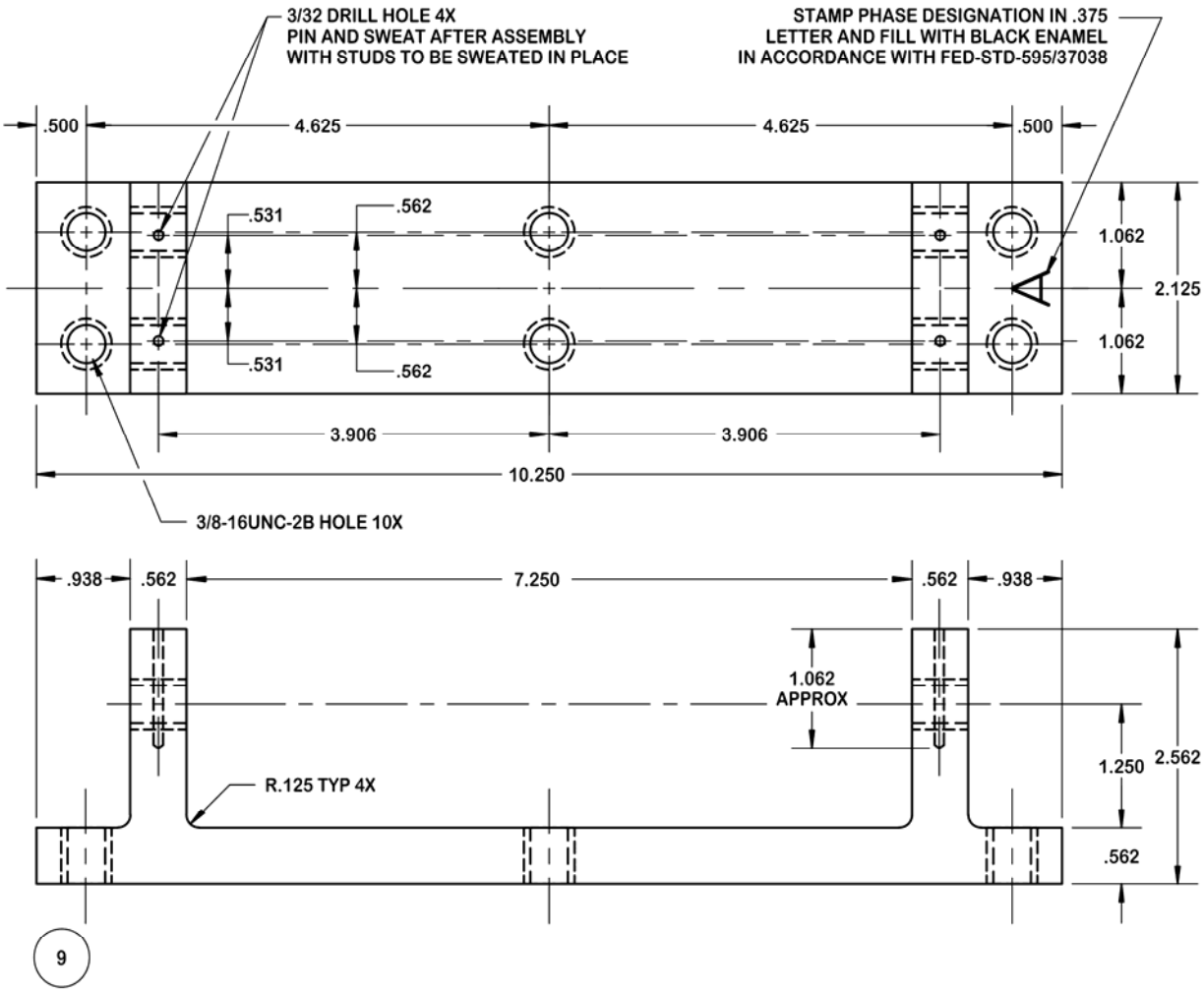


FIGURE 6. Terminal bus assembly - phase "A".

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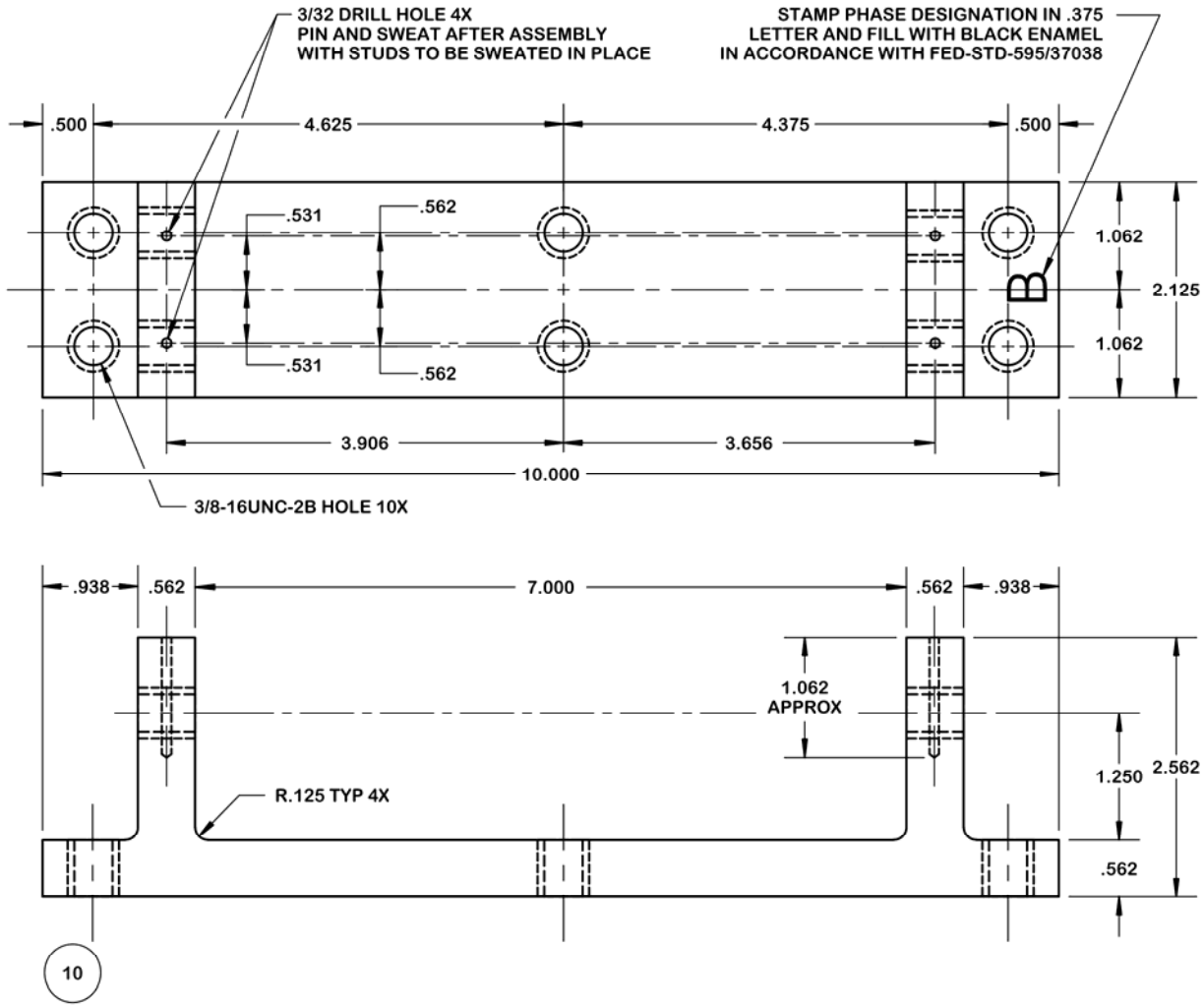


FIGURE 7. Terminal bus assembly - phase "B".

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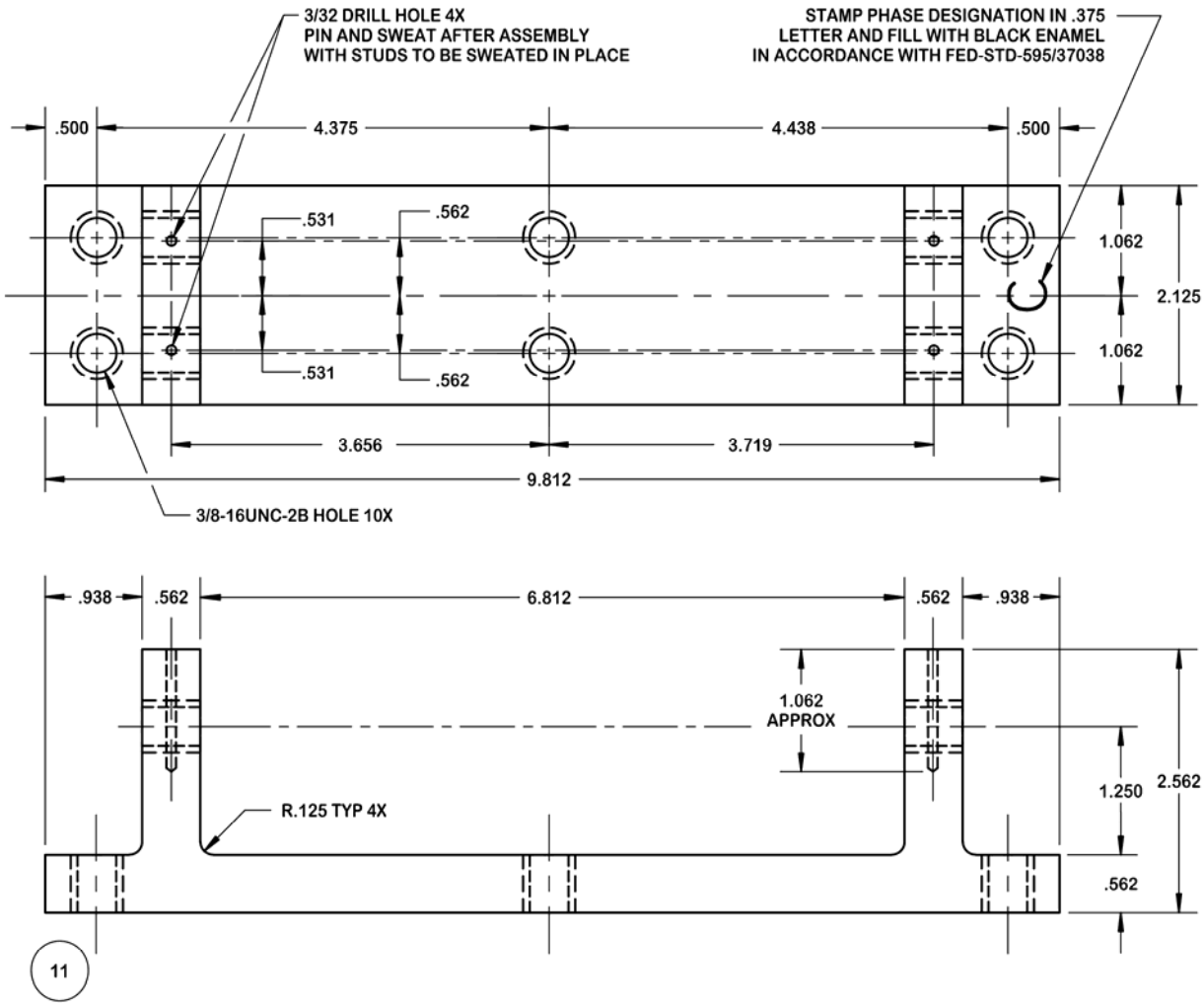


FIGURE 8. Terminal bus assembly - phase "C".

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy - SH
DLA - GS

Preparing Activity:
DLA - GS4

(Project 5940-2011-057)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at https://assist.daps.dla.mil/.

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